

The Human Genome

CS-3553 Assignment - 1

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1 1. Motivation

Modern technology and wide access to resources has helped most of the world know what the human genome is and why we use it. However, the deeper we study about this topic, the more curious one becomes about what was the actual motivation behind developing the Human Genome project. So this section is going to talk all about the motivation behind this evolution and why it has helped scientists and researchers advance within the field of biology.

During the late 1980s, scientists has discovered that certain techniques under recombinant DNA, had made it plausible to study individual genes and locate them. This progress, however, was only gradual. At that time, out of around 100,000 human genes, about 1,500 had been mapped, only 15 completely sequenced and around 500 partially sequenced. After calculation, this came out to be 1 percent of the 3 billion base pairs that had been read. The biggest issue, however, was that scientists were studying genes one-by-one which led to discovering diseases at the same pace. Some of these diseases include sickle cell anemia, cystic fibrosis and huntington [1]. This was one of the major issues leading upto to the study of the human genome. Since, if all the labs around world had a full map of the human DNA, they all would have a common starting point and this would in-turn lead to faster studies and discoveries about the human body.

Another reason that contributed towards the development of the human genome was the lack of cure/ treatment for genetic diseases [1]. During the same time (late 1980s), there were as many as 4,000 known genetic diseases. Most of these, however, had no remedy. Using the universally used human DNA map, doctors were able to detect 30 of these genetic diseases before birth, but still didn't guarantee a cure. Also, 30 out of a total of 4,000 meant that there was a lot of work left to do. Right around this period, scientists also found out that genes could be playing a huge role in commonly occouring human diseases like cancer, alzheimer's and heart disease.

The last reason, but not the least was funding and politics. The United States' government viewed the biotechnological industry one of the most fast-paced and evolutionary industries during the time. And since, economically competing countries like Japan had already started heavily investing into human gene research, there came a point when it was was inevitable for the US government to invest into the industry and help the grow the biotechnological advancement across the globe [1].

Roadblocks: There were a few big roadblocks that came into play before the US government went ahead with this evolutionary investment into biotechnology. Some

of them were - 1. The estimated investment was substantially large. The department of energy had quoted it to be one centrally run gene project that required anywhere between a few hundred million to a billion dollars. 2. Scientists argued that building better and stronger machines first, would help quicken the project and reduce the investment around 10-15 times. 3. There were quite a few biologists around the world that protested such an investment as it would take away smaller funded research projects handled by the National Institute of Health. Towards the later years, the National Academy of Sciences came up with a solid idea that help advance this decision and make the investment. They suggested to dedicate 3 billion dollars over the span of 15 years, and to begin this centrally run project by mapping everything they currently have about the human DNA, improving biological tools and spread their work across the world so that other countries could contribute towards the advancement as well [1].

Once this plan was finalized during the year 1988, the study of human genome project began. Their major goal was to study the unknown. So apart from just the human DNA, they divided their study so as to be able to study about smaller organisms like bacteria, yeast, etc. [2]

2 Milestones

2.1 Key Milestones

Now that we've understood why the study and research of the human genome began, let's look at some of the major milestone achieved all over the world in the biotechnological field right from very beginning (1980s). This section gives us a large scale overview of research discoveries, contributions made by major institutions and what each of them led to. It helps get a clear snapshot of the path the industry took to reach to modern day biotechnological advancements.

1986-1988: (a) The Department of Energy's (DOE) proposal to the US government about the large-scale investment. (b) The Academy of Sciences' report to the government. (c) The National Institute of Health (NIH) setting up it's genome head-quarters during 1988. [1,2]

1990: (a) The official beginning of the human genome project. (b) Launched alongside the project was a program to study the social, legal and ethical issue of the project and the industry. [2]

Early 1990s: (a) Scientists and researchers built a universally used human DNA map. [4]

1996: (a) The Bermuda conference/meeting. This was the meeting where it was accepted to share all DNA information publicly and as soon as discoveries had been made. [2]

1995-1999: (a) Beginning of the sequencing test-runs. (b) The finished study of the Chromosome 22. This was the first ever sequence to be finished. [4]

2000: (a) A rough draft of the human genome was published. This draft, however, had more than 150,000 estimated gaps in the study. [2]

2001: (a) The final draft was published by 20 research centers around the globe. These centers were located across 6 different countries. [3]

2003: (a) An amplified version of the draft was re-published. This version covered 92 percent (2 percent higher than the first draft), and had lower than 400 current gaps. This was a huge advancement covering most major gaps. [2]

2022: (a) The final few gaps had been filled in by latter researchers. This marked the publication of the first complete sequence worldwide. [2]

2.2 Methodological Challenges

Several multiple challenges had arisen during the course of this project. From biological errors, to political roadblocks to financial misuse. As recorded, these were some of the major methodological hurdles: 1. Limited Reading Methodology - The entirety of the project used one reading method to maintain consistency and convenience. The method used is called Sanger Sequencing. The issue with this tool was it showed signs of struggle with repetitive DNA sequences. [2]

2. Speed - On record, there were around 300,000 DNA chunks had to be scanned and fingerprinted in a single year [4]. This led to the project requiring a design team that processed 10 times more samples that it did earlier. This meant they were sampling 20,000 chunks a week.

3. Gaps in the Data - A lot of data used during the process had mismanaged data. While some of the entries had a wrong label, the other were just completely wrong data. What helped here, was using the map. However, even after that, there were more than a 1,000 gaps left in the map. [4]

4. Large gene size - At the time, the human gene was more than 20 times larger than anything read before [4]. And more than half of this were sequences of repeated DNA [3]. This required the machines to break the gene down into tiny pieces and then fit them back together for the outcome to make sense.

5. Repeated DNA - The repetitiveness of DNA sequences caused multiple smaller sized chunks to have the similar fingerprint. This made it difficult for applicants to differentiate and recognize one from the rest. Sorting out this was the largest consumer of time. [4]

2.3 Software Solutions Used

Before discussing the key softwares used as part of this huge project, some highly important names to note here are those of Maynard Olson and Ewan Birney [5]. It was Maynard's work on the yeast mapping in coherence with Ewan's work behind building the software for the project that led to the fast-paced evolution and the development of the project.

The relevant software solutions used are divided by the purpose they served. (i) Data Storage and Sharing (databases/websites): AceDb, GenBank, UCSC Genome Browser, Ensembl. [3,4]

(ii) Fingerprint Generation and Mapping: ePCR and in silico digests. [4]

(iii) Join chunks into a single genome: GigAssembler. [3]

(iv) Translation from Raw Signals to DNA letters and estimate confidence scores: PHRED and PHRAP. [3]

(v) Image reading from gel pictures and sorting repeated DNA sequences automatically: Image and FPC. [4]

3 Impact and Legacy

The impact this one centrally run project has created is massive and results show it. The legacy of such a wide-spread and carefully developed gene project is yet to be witnessed as we observe researchers and biologists all around the globe leveraging this study to make newer discoveries and deeply interesting finds.

If needed to summarize the benefits and impact of this investment into the human genome project, here are the major ones: (a) Substantially Cheaper DNA Reading: The generation of the first draft including the finished study took around 450 million dollars [6]. Ever since the first publication in the early 2000s, further funding has led to development of faster and more efficient machines. Likewise, the production of a single genome fell from 95 million dollars in 2001 to 343,000 in 2008 to as low as 525 dollars as of 2022 [7]. This price drop alone gave rise to much rapid advancement in the field.

(b) The Reveal of a Bigger Picture of the Human DNA: The initial draft of the project had found about 40,000 total genes out of 100,000 expected ones [3]. This number has grown much higher since then. The availability and access of cheaper DNA reading has hugely impacted this. This discovery gave rise to a much larger picture of the human DNA and how genes govern various aspects of the human body and health. This same discovery also was a starting point into the study of why people differ from one-another.

(c) The Study of Differences among Humans: The human genome project led to a later developed project called the HapMap project. The basis of this study was to analyze genomes for DNA variations and how those differences relate to various prominent diseases [8]. A later connected project called the 1000 Genomes project had been developed, which found greater than 88 million variations in DNA sequences from the several human genes taken into sample [9]. This however, wasn't a very successful study as most links found among the genomes were only tiny effects. [8]

(d) The Junk DNA debate: The ENCODE project, which is highly prominent even during recent times, reports that over 80 percent of the human genome does something, but most disease-causing variations exist outside this 80 percent [8]. The debate over this topic is with the scientists showing a strong disapproval of the 80 percent. Either of the co-existing arguments are yet to be proven.

(e) The Medical Field: The human genome project has created some of the most valuable advancements in the healthcare industry [7]. (1) Cancer. In the treatment of cancer, doctor's are now able to read a tumour's DNA to choose a drug that specifically targets it. This increased the efficacy of the drugs used. (2) Drug safety. Doctors can determine whether the usage of a particular drug would cause cure or harm to a particular person. This helps with drug prescription. (3) Pregnancy. A simple blood test from the mother, can help test the baby's health and help detect early diseases in the baby. (4) Diagnosis of Unusual Diseases. Doctors can use the help of DNA tests to explain unusual medical conditions in kids.

(f) Open Sharing of Data: The Bermuda meeting contributed a lot towards the open sharing and wide-spread access of DNA data. Ever since the meeting had convened,

the access to all the data used during the human genome project is available to all biotechnological labs all over the world. This helps creating an equal playing ground for researchers as well as scientists. [2]

There were numerous more impacts that were created by the human genome project up to the present day [2]. The list goes on, however this mentioned were some of the most beneficial as regarded by various professionals around the world. The additional outcomes also include advancement in national law and ethics in the United States, huge economical upside [10], and a whole new lens to view the biotechnological industry through. [2]

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